A Theory of Slack

How Economic Slack Shapes Markets, Business Cycles, and Policies

Pascal Michaillat

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CHAPTER 18.

Fiscal policy

In chapter 17 we saw how monetary policy should respond when the economy is not at full employment. In various circumstances, it is either not possible or not desirable for monetary policy to eliminate the unemployment gap entirely. In that case, it is natural to wonder: can monetary policy be complemented by fiscal policy? As we see in this chapter, the answer to this question is yes.

The first case when fiscal policy might be useful is when monetary policy is impotent. This occurs at the zero lower bound on the nominal interest rate. Once the zero lower bound is reached, interest rates can no longer be reduced, so monetary policy can no longer be used to prop up the economy.

The second case when monetary policy cannot be used is under monetary unions. In a monetary union, monetary policy is decided at a higher level and cannot be controlled by the local government. One example of a monetary union is the United States; here, monetary policy is controlled by the Federal Reserve and not by individual states. States may be subject to local recessions that are not synchronized with national recessions and that are therefore not tackled by the Federal Reserve (Owyang, Piger, and Wall 2005). But states are able to determine how much they want to spend, how much they want to tax, and they can even issue bonds. Hence states are in complete control of their fiscal policy, and can use it to tackle local business cycles.

18.1. Beveridge curve

To study optimal fiscal policy, we follow the same approach we used to study the unemployment gap in chapter 13 and optimal monetary policy in chapter 17. We specify a generic framework and derive a sufficient-statistic formula for the optimal fiscal policy.

A central assumption for the analysis, which we have made before, is that there is a Beveridge curve, $v(u)$, so that we always have some level of unemployment in the model and we are able to have a sufficient-statistic formula for u^*. This means that we are easily able to compute the unemployment gap.

18.2. Public versus private employment

The key novelty in the analysis of fiscal policy comes from work on the provision of public goods by Samuelson (1954, 1955). This theory shows that public goods should be provided to the point where the marginal rate of substitution between public and private consumption equals their marginal rate of transformation.[1]

The idea is that there are two types of goods, public and private, and two types of workers. Some workers are employed directly by households to provide private goods, and some are employed by the government to produce public goods. We assume that the government provides g public goods by recruiting and employing workers, and that the public goods benefit everyone. These public goods enter the household's utility function. Of course, there is a resource cost to provide public goods, since public workers do not work in the private sector. There are only a finite number of workers, so there is a tradeoff and the government can hire more or less workers to stabilize the economy.

Let us now try to look at the composition of the labor force. We take the size of the labor force as exogenous so that the number of workers hired by the government has no effect on the size of the labor force. It is therefore not considered in our welfare analysis. It is important to look at how the labor force is composed because the number of workers in different sectors immediately determines, through a production function, the number of services they produce, and this determines welfare.

We have four types of workers. First, a share u of the workers is unemployed, so they do not contribute to welfare (assuming that there is no home production by unemployed workers). Next, a share v of workers who are employed and recruiting. Both households and the government recruit in a similar way and need recruiters to fill vacancies and hire workers. The Beveridge curve, $v(u)$, links the number of recruiters to the number of unemployed workers. Note that workers do not produce consumable services while recruiting.

[1] See Kreiner and Verdelin (2012) for a survey of the literature on the optimal provision of public goods.

We also have a share $1 - (u + v)$ of workers who are employed and producing services. There are two types of employers: private employers (households) and public employers (the government). A share c of the labor force is employed by households and produces private services, while a share g of the labor force is employed by the government and produces public services. Note that $c+g = 1-(u+v)$ represents total productive employment. Thus,

$$c = 1 - (u + v) - g,$$

where c is private productive employment, $u + v$ is nonproductive use of labor (recruiting and job-seeking), and g is public productive employment (which is what creates distortion, since resources are being depleted from the private sector). This tradeoff between c and g is at the heart of the Samuelson framework.

18.3. Marginal rate of substitution between public and private goods

A key aspect of the Samuelson framework is that households derive utility from both public and private services. Thus, both c (the share of private productive workers) and g (the share of public productive workers) enter the representative household's utility function, $\mathcal{U}(c, g)$. We assume that $\mathcal{U}$ is strictly increasing in c and g, and that it is strictly concave so that our optimization problem is well-behaved.

An important term that features in our optimization problem is the marginal rate of substitution between public and private services, $MRS_{g,c}$. This tells us how the household values public services in terms of private services:

$$MRS_{g,c} = \frac{d\mathcal{U}/dg}{d\mathcal{U}/dc} > 0.$$

We assume that $MRS_{g,c}$ is strictly decreasing in g/c so that when there are more public services relative to private services, the value of a public service relative to a private service decreases. We also assume that $MRS_{g,c}(g/c = 0) > 1$, meaning that when there are no public services, a public service is more valuable than a private service. This is required so that it is desirable to provide some public goods (and so we can get an interior solution to our optimization problem).

We are now in a position to introduce the first sufficient statistic that appears in the formula for optimal government spending: the elasticity of substitution between public and private services, ϵ. It is defined by the equation:

$$\frac{1}{\epsilon} = -\frac{d\ln\left(MRS_{g,c}\right)}{d\ln(g/c)}. \tag{18.1}$$

Since $MRS_{g,c}$ is decreasing in g/c, we see that $\epsilon > 0$. We have three different possibilities.

First, when $\epsilon < 1$, public and private services are gross complements. Second, when $\epsilon = 1$, public and private services are independent. Last, when $\epsilon > 1$, public and private services are gross substitutes.

Let us now consider the extreme case where public and private services are perfect complements ($\epsilon \to 0$). In this case, we have a Leontief utility function, where

$$\mathcal{U}(c, g) = \min(c, g).$$

Another extreme case is when $\epsilon = 1$ so that public and private services are independent. Then, we would have a Cobb-Douglas utility function:

$$\mathcal{U}(c, g) = c^{1-\gamma} g^{\gamma}.$$

The last extreme case is when $\epsilon \to \infty$, and public and private goods are perfect substitutes. Here, our utility function would be linear:

$$\mathcal{U}(c, g) = c + g.$$

18.4. Unemployment multiplier

The other important sufficient statistic that shows up in the formula for optimal government spending is the unemployment multiplier, m. It is the decrease in the unemployment rate (in percentage points) when the share of the labor force devoted to public production increases by 1 percentage point. We define it by the equation:

$$m = -\frac{du}{dg}.$$

In our model, public employment affects the unemployment rate because government spending affects aggregate demand, which affects tightness. This directly determines the unemployment rate through the aggregate supply (the Beveridge curve).

There is an extreme case when $m = 0$ and the unemployment rate does not respond to public spending. Although this is a possibility, empirical evidence from the United States suggests that the unemployment rate drops when public spending goes up. Thus, a more realistic case is $m > 0$. Note that our framework also allows for m to be negative but, again, this is not what the data shows.

18.5. Effects of government spending on welfare

We now formalize the social planner's problem. The planner's goal is to maximize welfare $\mathcal{U}(c, g)$ subject to the resource constraint $c = 1 - (u + v) - g$. Substituting private consumption from the welfare function, the planner's problem simplifies to

$$\max_g (\mathcal{U}(1-(u+v)-g,g)).$$

Here, we see the tradeoff that the government faces. In addition to our resource constraint, we also have a stabilization assumption: government spending affects the unemployment rate ($u(g)$) as well as the recruiting rate ($v(g)$). Thus, the social planner's problem is:

$$\max_g \mathcal{U}(1-(u(g)+v(u(g)))-g,g).$$

To better understand the tradeoffs and how public spending affects welfare, we take the derivative of social welfare with respect to g:

$$\frac{d\mathcal{U}}{dg}+\frac{d\mathcal{U}}{dc}\left[-u'(g)-v'(u)\cdot u'(g)-1\right],$$

which leads to

$$\frac{d\mathcal{U}}{dg}-\frac{d\mathcal{U}}{dc}+\frac{d\mathcal{U}}{dc}\cdot\left[-u'(g)\right]\cdot\left[1+v'(u)\right].$$

Here, we see all the ways through which public spending affects welfare. The first term tells us that having more public services increases welfare, and the second term tells us that if we have more public workers, there are fewer private workers and therefore less welfare (Samuelson 1954). The third term is the stabilization aspect—it shows that g affects u.

18.6. Optimal government spending

We now determine the optimal amount of government spending g, which maximizes $\mathcal{U}(c(g),g)$. We assume that $\mathcal{U}(c(g),g)$ is a well-behaved function of g so that it admits a unique extremum which is an interior maximum. The first-order condition is therefore necessary and sufficient to find the solution of the social planner's problem. Taking the first-order condition, we get

$$0=\frac{\partial\mathcal{U}}{\partial g}-\frac{\partial\mathcal{U}}{\partial c}+\frac{\partial\mathcal{U}}{\partial c}\cdot\left[-u'(g)\right]\left[1-(-v'(u))\right],$$

which gives

$$\frac{\partial\mathcal{U}}{\partial c}=\frac{\partial\mathcal{U}}{\partial g}+\frac{\partial\mathcal{U}}{\partial c}\cdot\left[-u'(g)\right]\left[1-(-v'(u))\right].$$

We rewrite this using our sufficient statistics that we defined earlier:

$$1=MRS_{g,c}+\left[-u'(g)\right]\cdot\left[1-(-v'(u))\right]$$

which then gives

$$1 = MRS_{g,c} + m \cdot \left[1 - (-v'(u))\right].$$

This is our first formula for optimal government spending, however we need to do a little more work to make it more explicit.

Before we do so, it is important to highlight two key things. First, note that the Samuelson rule, ($MRS_{g,c} = 1$), appears here. There is an extra term in our formula, however, which appears because we are no longer in a neoclassical model—we have an inefficient level of slack. Thus, we need a correction/stabilization term since we are in an economy that is not stabilized and has productive inefficiencies.

18.7. Samuelson rule

As we have seen, the Samuelson rule, $MRS_{g,c}(g/c) = 1$, is a key element in our optimal public spending formula. Using this, we define Samuelson spending, $(g/c)^*$, which is the amount of public spending implied by the Samuelson rule. This is the optimal level of public spending in a neoclassical model. Taking the first-order approximation of $MRS_{g,c}$ around $(g/c)^*$, we get:

$$MRS_{g,c} \approx MRS_{g,c}((g/c)^*) + \frac{dMRS_{g,c}}{dg/c} \cdot \left[g/c - (g/c)^*\right].$$

Given that $MRS_{g,c}((g/c)^*) = 1$, and given the definition of the elasticity of substitution (18.1), this expression simplifies to:

$$1 - MRS_{g,c} = \frac{1}{\epsilon} \cdot \frac{g/c - (g/c)^*}{(g/c)^*},$$

where ϵ is the elasticity of substitution between public and private services and $\frac{g/c-(g/c)^*}{(g/c)^*}$ is the departure from Samuelson spending—the amount of stimulus spending.

18.8. Stabilization term

In addition to the Samuelson term in our optimal government spending formula, we also have a stabilization term:

$$m\left[1 - (-v'(u))\right],$$

where $m = -du/dg$ is the unemployment multiplier and $-v'(u)$ is the slope of the Beveridge curve. Since $-v'(u)$ is more difficult to interpret, let us rework it. Taking the first-order approximation of $-v'(u)$ around u^*, we get:

$$-v'(u) \approx -v'(u^*) - v''(u^*)(u - u^*).$$

Recall that u^* is the unemployment rate that minimizes $u + v(u)$, so $1 + v'(u^*) = 0$, which gives

$$-v'(u^*) = 1.$$

Furthermore, since we are assuming that $v(u)$ is a hyperbola, we work out that

$$v''(u^*) = -v'(u^*)\frac{2}{u^*} = \frac{2}{u^*},$$

since $-v'(u^*) = 1$. We use these results to simplify the first-order approximation:

$$-v'(u) \approx 1 - \frac{2}{u^*}(u - u^*)$$

so that

$$1 - (-v'(u)) = 2\frac{u - u^*}{u^*}.$$

Here, we see that the difference between 1 and the slope of the Beveridge curve is equal to twice the relative deviation of the unemployment rate compared to the efficient level of unemployment. This relationship allows us to introduce the unemployment gap into our optimal government spending formula.

18.9. Implicit formula for optimal stimulus spending

By combining the previous results, we obtain the following formula for optimal government spending:

$$\frac{g/c - (g/c)^*}{(g/c)^*} = 2\epsilon \cdot m \cdot \frac{u - u^*}{u^*}. \tag{18.2}$$

A first insight from formula (18.2) is that the Samuelson rule, which was originally derived in a neoclassical model, remains valid in a model with unemployment as long as unemployment is efficient. Indeed, when unemployment is efficient, $u = u^*$, so the formula says that $g/c = (g/c)^*$: government spending must be at the Samuelson level.

A second implication of the formula is that it is never optimal to deviate from the Samuelson rule enough to eliminate the unemployment gap—it is only optimal to reduce the unemployment gap, when it exists. If the government adjusts spending to stabilize the economy, $g/c \neq (g/c)^*$, so the left-hand side of the equation is nonzero. If the spending eliminates the unemployment gap, $u - u^* = 0$, so the right-hand side of the equation is zero. So the two sides of the equation cannot be equal, which tells us that eliminating the unemployment gap cannot be optimal.

To understand this result, we must bear in mind that, even though fiscal policy can be used for stabilization, it creates some distortion. When the government decides to increase spending by providing more public goods, it distorts the provision of goods in the economy away from private goods. While private goods can be substituted for public goods—public schools and hospitals can replace private schools and hospitals—they are not perfect substitutes. Due to these distortions, fiscal policy is not able to bring the economy all the way to the efficient allocation. Therefore, it is not optimal to stabilize the economy perfectly with fiscal policy alone: it is not optimal to close the unemployment gap entirely.

18.10. Explicit formula for optimal stimulus spending

Our sufficient-statistic formula (18.2) is only implicitly defined because u depends on g/c itself. The next step is to rewrite this formula to make the dependence explicit so that we actually quantify optimal stimulus spending.

To do this, we express $\frac{u-u^*}{u^*}$ as a function of $\frac{u_0-u^*}{u^*}$ and $\frac{g/c-(g/c)^*}{(g/c)^*}$. Taking the first-order approximation of $\frac{u-u^*}{u^*}$ around the initial situation, we get:

$$\frac{u-u^*}{u^*} = \frac{u_0-u^*}{u^*} + \frac{1}{u^*} \cdot \frac{du}{d(g/c)} \cdot (g/c - (g/c)^*),$$

which implies

$$\frac{u-u^*}{u^*} = \frac{u_0-u^*}{u^*} + \frac{1}{u^*} \cdot \frac{du}{d\ln(g/c)} \cdot \frac{g/c-(g/c)^*}{(g/c)^*}.$$

Computing $du/d\ln(g/c)$, we get:

$$\frac{du}{d\ln(g/c)} = \frac{du}{dg} \cdot \frac{dg}{d\ln(g/c)} = -m \cdot \frac{dg}{d\ln(g/c)}$$

We rework $dg/d\ln(g/c)$ using the fact that $c = 1-(u+v)-g$. Then,

$$\frac{dc}{dg} = -\frac{du+v}{dg} - 1.$$

At the point $(u^*, (g/c)^*)$, we have

$$\frac{du+v}{dg} = 0,$$

so that, necessarily,

$$\frac{dc}{dg} = -1.$$

With this result, we compute $d\ln(g/c)/dg$:

$$\frac{d\ln(g/c)}{dg} = \frac{1}{g/c} \cdot \frac{d(g/c)}{dg} = \frac{1}{g/c}\left(\frac{1}{c} - \frac{g}{c^2} \cdot \frac{dc}{dg}\right).$$

At $(g/c)^*$,

$$\frac{d\ln(g/c)}{dg} = \frac{c^*}{g^*}(\frac{1}{c^*} + (\frac{g}{c^2})^* \cdot \frac{dc}{dg}) = \frac{1}{g^*} + \frac{1}{c^*}.$$

We now go back and compute $du/d\ln(g/c)$:

$$\frac{du}{d\ln(g/c)} = -m \cdot \frac{1}{\frac{1}{g^*} + \frac{1}{c^*}}.$$

We rewrite this as:

$$\frac{du}{d\ln(g/c)} = -\frac{m}{2} \cdot \frac{2}{\frac{1}{g^*} + \frac{1}{c^*}},$$

so that the second fraction is the harmonic mean of g^* and c^*. Plugging $du/d\ln(g/c)$ back into the first-order approximation, we get:

$$\frac{u - u^*}{u^*} = \frac{u_0 - u^*}{u^*} - \frac{m}{2u^*} \cdot \frac{2}{\frac{1}{g^*} + \frac{1}{c^*}} \cdot \frac{g/c - (g/c)^*}{(g/c)^*}.$$

We have now expressed the unemployment gap using the initial unemployment gap, our sufficient statistics, and stimulus spending. We now plug this into the right-hand side of our implicit formula to make it explicit:

$$\frac{g/c - (g/c)^*}{(g/c)^*} = 2\epsilon m \cdot \frac{u_0 - u^*}{u^*} - \epsilon m^2 \cdot \frac{1}{u^*} \cdot \frac{2}{\frac{1}{g^*} + \frac{1}{c^*}} \cdot \frac{g/c - (g/c)^*}{(g/c)^*},$$

which then gives

$$\left(1 + \epsilon m^2 \cdot \frac{1}{u^*} \cdot \frac{2}{\frac{1}{g^*} + \frac{1}{c^*}}\right) \cdot \frac{g/c - (g/c)^*}{(g/c)^*} = 2\epsilon m \cdot \frac{u_0 - u^*}{u^*}.$$

Then, our explicit formula for optimal stimulus spending is:

$$\frac{g/c - (g/c)^*}{(g/c)^*} = \frac{2\epsilon m}{1 + z\epsilon m^2} \cdot \frac{u_0 - u^*}{u^*}, \tag{18.3}$$

where the constant z is given by

$$z = \frac{1}{u^*} \cdot \frac{2}{\frac{1}{g^*} + \frac{1}{c^*}}.$$

Recall that m is the unemployment multiplier, and ϵ is the elasticity of substitution between g and c.

TABLE 18.1. Optimal deviation from Samuelson rule when unemployment is inefficient

	Unemployment multiplier		
Unemployment gap	$m < 0$	$m = 0$	$m > 0$
$u > u^*$	$g/c < (g/c)^*$	$g/c = (g/c)^*$	$g/c > (g/c)^*$
$u = u^*$	$g/c = (g/c)^*$	$g/c = (g/c)^*$	$g/c = (g/c)^*$
$u < u^*$	$g/c > (g/c)^*$	$g/c = (g/c)^*$	$g/c < (g/c)^*$

The optimal deviations from the Samuelson rule are derived from equation (18.2).

18.11. Properties of optimal stimulus spending

In this section we describe how the optimal amount of stimulus spending, given by formula (18.3), depends on the sufficient statistics: unemployment gap, elasticity of substitution between public and private goods, and unemployment multiplier.

18.11.1. Optimal deviation from the Samuelson rule

Formula (18.3) first describes qualitatively how optimal government spending deviates from Samuelson spending. There are several scenarios to consider (table 18.1).

Overall, if the unemployment multiplier is zero ($m = 0$), or the unemployment gap is zero ($u_0 = u^*$), optimal stimulus spending is zero ($g/c = (g/c)^*$). Otherwise, optimal public expenditure deviates from the Samuelson rule to partially fill the initial unemployment gap. Thus, optimal public expenditure satisfies the Samuelson rule only if the unemployment multiplier is zero or if the unemployment gap is zero. In all other situations, optimal public expenditure deviates from the Samuelson rule.

Consider first a positive unemployment multiplier ($m > 0$). If the unemployment gap is positive ($u_0 > u^*$), optimal stimulus spending is positive ($g/c > (g/c)^*$) but does not completely fill the unemployment gap ($u > u^*$). If the unemployment gap is negative ($u_0 < u^*$), optimal stimulus spending is negative ($g/c < (g/c)^*$) but does not completely eliminate the unemployment gap ($u < u^*$).

On the other hand, when the unemployment multiplier is negative ($m < 0$), optimal stimulus spending is negative if the unemployment gap is positive, and optimal stimulus spending is positive if the unemployment gap is negative.

This result is simple to understand. At the Samuelson rule, an increase in public expenditure has no first-order effect on welfare when we ignore its effect on unemployment. Now, when the unemployment multiplier is positive, an increase in public expenditure lowers unemployment; and when the unemployment gap is positive, unemployment is inefficiently high, so lowering unemployment raises welfare. Hence, overall, an increase in public expenditure generates a positive first-order effect on welfare. It is therefore optimal to increase public expenditure above the Samuelson rule. Further, since the unemployment gap measures the welfare gain from reducing unemployment, optimal stimulus spending is increasing in the unemployment gap.

The general pattern is that optimal public expenditure deviates from the Samuelson rule to partially fill the initial unemployment gap. To understand these results, imagine that public expenditure satisfies the Samuelson rule, the unemployment multiplier is positive, and unemployment is inefficiently high. Keeping total consumption constant, increasing public consumption reduces private consumption one-for-one. Since the marginal utilities of public and private consumption are equal at the Samuelson rule, the increase in public consumption has no first-order effect on welfare so far. Now, since the unemployment multiplier is positive, increasing public consumption lowers unemployment; and since unemployment is inefficiently high, reducing unemployment raises total consumption. Hence, through its effect on unemployment, the increase in public consumption raises welfare. It is therefore optimal to increase public consumption above the Samuelson rule, and thus reduce the unemployment gap.

Why is it not optimal to completely fill the unemployment gap? If the government did that, we would reach a situation where increasing public consumption reduces private consumption one-for-one (since crowding out is one-for-one when the unemployment gap is zero), but extra public consumption is less valuable than extra private consumption (since public consumption is above the Samuelson level). The situation is clearly suboptimal: welfare can be improved by reducing public consumption.

18.11.2. Role of the unemployment gap

We now look at the role of the initial unemployment gap on the design of optimal stimulus spending. We see in formula (18.3) that optimal stimulus spending is larger when the unemployment gap is larger. This makes sense because a larger unemployment gap means there is more inefficient slack, which must be tackled with more spending.

18.11.3. Role of the unemployment multiplier

Next, we look at the role of the multiplier. The relationship between the unemployment multiplier and stimulus spending is a little more interesting—the function is hump-shaped. Specifically, at $m = 0$, there is no stimulus spending. It then increases and reaches a peak value at m^*, which we compute by taking the derivative of our explicit formula $2\epsilon m/[1 + z\epsilon m^2]$ with respect to m and setting it equal to zero. After some reshuffling, we get:

$$m^* = \frac{1}{\sqrt{z\epsilon}}.$$

This is the value of m that maximizes stimulus spending. It then decreases with m when $m > m^*$.

What is the intuition behind the hump shape? When government spending is optimal, the marginal social cost from consuming too many public services and too few private services equals the marginal social value from reducing unemployment. This marginal social value is determined by two factors: the current unemployment multiplier, which measures how much unemployment can be reduced by additional expenditure, and the current unemployment gap, which measures the social value from lower unemployment. For a given amount of stimulus spending and a given initial unemployment gap, a larger initial multiplier has conflicting effects on the two factors: it means a larger current multiplier (a higher marginal social value) but a smaller current unemployment gap (a lower marginal social value). The first effect advocates for more spending, the second for less spending.

It turns out that for small multipliers, the first effect dominates, so optimal stimulus spending is increasing in the multiplier; for large multipliers, the second effect dominates, so optimal stimulus spending is decreasing in the multiplier. In fact, for very large multipliers, it becomes optimal to nearly entirely fill the unemployment gap. Naturally, less spending is required to fill the gap when the multiplier is larger, so optimal stimulus spending is decreasing in the multiplier.

Hence, the bang-for-the-buck logic often used in policy discussions—the view that a larger multiplier entails a larger stimulus spending—is not generally valid. Stimulus skeptics usually believe in small multipliers and infer that stimulus spending should be small in slumps. Similarly, stimulus advocates usually believe in large multipliers and infer that stimulus spending should be large in slumps. We have just seen that this bang-for-the-buck logic holds for small multipliers but not for large ones: a large multiplier is not a justification for a large stimulus. Instead, since the relationship between multiplier and optimal stimulus spending is hump-shaped, optimal stimulus spending is similar for some small and large multipliers.

18.11.4. Role of the elasticity of substitution

The policy debate on stimulus spending often revolves around unemployment gaps and multipliers. Formula (18.3) confirms that optimal stimulus spending is indeed a function of the unemployment gap and a multiplier—the unemployment multiplier. Yet, these statistics are not sufficient to measure the effect of public expenditure on welfare because an increase in public expenditure also modifies the composition of households' consumption. Consequently, optimal stimulus spending also depends on the elasticity of substitution between public and private consumption; this statistic should probably play a more prominent role in the policy debate.

What exactly is the role of the elasticity of substitution? We see that stimulus spending is increasing in ϵ. When there is higher substitutability, the stimulus package is larger. At $\epsilon = 0$, there is no stimulus spending, and at $\epsilon \to \infty$, we have that

$$\frac{g/c - (g/c)^*}{(g/c)^*} = \frac{2}{zm} \frac{u_0 - u^*}{u^*}.$$

What is the intuition behind the case $\epsilon \to 0$? In this case, additional public services have zero value: additional public workers dig and fill holes in the ground. Then, optimal stimulus spending is zero, irrespective of the unemployment rate and multiplier. Intuitively, above the Samuelson level, public consumption has no value, but it crowds out private consumption; therefore, it cannot be optimal to provide more public consumption than the Samuelson level.

But for any $\epsilon > 0$, public services have some value, and optimal stimulus spending is nonzero as long as the economy is inefficient and the multiplier is nonzero. This result clarifies the link between usefulness of public expenditure and optimal stimulus spending. A concern of stimulus skeptics is that additional public expenditure could be wasteful. It is true that when the elasticity of substitution between public and private consumption is zero, public expenditure should remain at the Samuelson level. But in the more realistic case where the elasticity of substitution is positive, some stimulus spending remains desirable in slumps.

18.12. Stabilization from optimal stimulus spending

We know that fiscal policy is less powerful than monetary policy because here, it is not optimal to completely fill the unemployment gap since fiscal policy creates distortions. The question, then, is: how much of the unemployment gap should we fill and how does that depend on our sufficient statistics? Recall our formula for the optimal government spending:

$$\frac{g/c - (g/c)^*}{(g/c)^*} = \frac{2\epsilon m}{1 + z\epsilon m^2} \cdot \frac{u_0 - u^*}{u^*}.$$

In addition to this, we also know how the unemployment gap depends on public spending and the initial unemployment gap. We showed that the resulting unemployment gap after government spending is given by:

$$\frac{u - u^*}{u^*} = \frac{u_0 - u^*}{u^*} - \frac{mz}{2} \cdot \frac{g/c - (g/c)^*}{(g/c)^*}.$$

We now combine the optimal stimulus package with the effect of the stimulus on unemployment to figure out how much of the unemployment gap is left once public spending is optimal:

$$\frac{u-u^*}{u^*}=\frac{u_0-u^*}{u^*}-\frac{mz}{2}\cdot\left(\frac{2\epsilon m}{1+z\epsilon m^2}\right)\cdot\frac{u_0-u^*}{u^*},$$

which gives

$$\frac{u-u^*}{u^*}=\left(1-\frac{z\epsilon m^2}{1+z\epsilon m^2}\right)\cdot\frac{u_0-u^*}{u^*},$$

and which finally leads to

$$u-u^*=\frac{1}{1+z\epsilon m^2}\cdot(u_0-u^*). \tag{18.4}$$

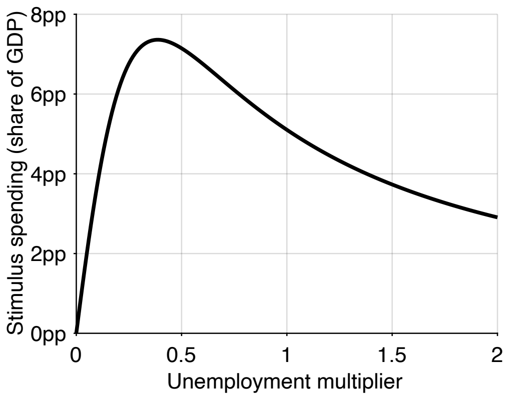
This key relationship tells us that the final unemployment gap (with optimal stimulus) is a share $\frac{1}{1+z\epsilon m^2}$ of the initial unemployment gap. Notice that the share is less than 1, so optimal stimulus spending reduces the original unemployment gap. But the share is strictly positive, so stimulus spending never eliminates the unemployment gap entirely—it only reduces it. We also see that the share is decreasing with ϵ—the final unemployment gap is smaller when public goods are more substitutable with private goods. Furthermore, the share is also decreasing with m—the final unemployment decreases when the multiplier is larger.

An interesting special case is that the final unemployment gap converges to 0 when $\epsilon\to\infty$. In this case, public consumption perfectly substitutes for private consumption, and optimal stimulus spending completely fills the unemployment gap. This result holds even if the multiplier is very small and public expenditure severely crowds out private consumption. Intuitively, public consumption and private consumption are interchangeable, so it is optimal to maximize total consumption, irrespective of its composition. This is achieved by completely filling the unemployment gap.

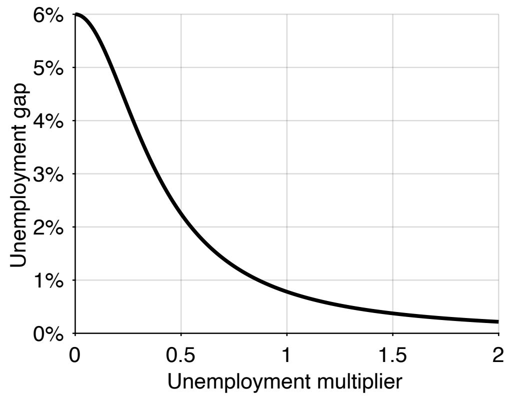
18.13. Application to the Great Recession

We now complement our theoretical results with a numerical application. We calibrate our sufficient-statistic formula and compute optimal stimulus spending at the onset of the Great Recession in the United States. This exercise illustrates how much optimal public expenditure may deviate from the Samuelson rule.

The motivation is that monetary policy hit the zero lower bound and became ineffective in 2009 (Board of Governors of the Federal Reserve System 2025), while the unemployment gap reached 5.9pp in the same year (figure 1). There was therefore a broad scope for fiscal policy in the form of government spending at the time. So according to our formulas, how much should the government have spent in 2009?



A. Optimal stimulus spending



B. Optimal unemployment gap

FIGURE 18.1. Optimal stimulus spending for an initial unemployment gap of 6pp

Optimal stimulus spending is given by the explicit sufficient-statistic formula (18.3). The optimal unemployment gap is given by formula (18.4).

18.13.1. Calibration of the sufficient statistics

We set the Samuelson level of spending to $(g/c)^* = 19.7\%$, which is its 25-year average. To construct g/c, we set g to the seasonally adjusted employment level in the government industry and c to the seasonally adjusted employment level in the private industry. Both series constructed are by the Bureau of Labor Statistics from the Current Employment Statistics survey. Over the 1990–2014 period, the average public expenditure satisfies $g/c = 19.7\%$.

To estimate the elasticity of substitution between public and private consumption, the empirical strategy is to isolate variations in the ratio of public-consumption price to private-consumption price and to assess their impact on the ratio of public consumption to private consumption. For example, if the consumption ratio stays constant in spite of secular variations in the price ratio, then the elasticity of substitution is about one. Applying this approach to US data, Amano and Wirjanto (1997, 1998) estimate elasticities of 0.9 and 1.56. Hence, we pick a midrange value for the elasticity of substitution: $\epsilon = 1$.

Next, we determine plausible values for the unemployment multiplier. The unemployment multiplier is estimated by measuring the percentage-point change in the unemployment rate when public expenditure increases by one percent of GDP. Monacelli, Perotti, and Trigari (2010, pp. 533–536) estimate a structural vector autoregression on US data and find unemployment multipliers between 0.2 and 0.6. Ramey (2013, pp. 40–42) also estimates structural vector autoregressions on US data with various identification schemes and sample periods. She finds unemployment multipliers between 0.2 and 0.5, except in one specification where the multiplier is 1. In sum, the average unemployment multiplier is estimated to be in the 0.2–1 range.

The multiplier entering our formula could be larger if multipliers are larger when unemployment is higher, as we find in slackish models (chapter 9). To account for the uncertainty about the exact value of the multiplier in bad times, we compute optimal stimulus spending for a range of unemployment multipliers: $0 < m < 2$.

The last step before using the formulas is calibrating the constant z. We calibrate it just as above: we set $(g/c)^* = 19.7\%$, $(g/y)^* = 16.5\%$, $(c/y)^* = 1 - (g/y)^* = 83.5\%$, $u^* = 4\%$, which implies $z = 6.67$.

18.13.2. Calculation with sufficient-statistic formulas

Figures 18.1A and 18.1B display optimal stimulus spending as a share of GDP ($g/y - (g/y)^*$), and unemployment rate under optimal stimulus spending.[2] Several observations stand out.

First, even with small multipliers, optimal stimulus spending is significant. For example, take a tiny multiplier of 0.05: optimal stimulus spending is 2.0 points of GDP. For a small multiplier of 0.1, optimal stimulus spending is 3.7 points of GDP. And for a modest multiplier of 0.2, optimal stimulus spending is 6.1 points of GDP.

Second, the multiplier warranting the largest stimulus is fairly modest. The largest stimulus, which is 7.4 points of GDP, occurs at a multiplier of 0.39.

Third, optimal stimulus spending is the same for small and large multipliers. For instance, optimal stimulus spending is the same, at 3.7 points of GDP, for a small multiplier of 0.1 and a large multiplier of 1.5. Of course the resulting unemployment rates are very different.

Fourth, for small multipliers, the unemployment gap barely falls below the initial value of 6pp although optimal stimulus spending is large. For example, with a multiplier of 0.1, the unemployment gap only falls by 0.4pp to 5.6pp. This is because public spending has little effect on unemployment when the multiplier is small.

Fifth, with a multiplier above 1, optimal stimulus spending almost brings back unemployment to its efficient level, so the unemployment gap almost vanishes. Indeed, when the multiplier is above 1, the remaining unemployment gap is less than 0.8pp. And when the multiplier reaches 2, the remaining unemployment gap is just 0.2pp.

Finally, we calculate optimal stimulus spending at the onset of the Great Recession using midrange values for the unemployment multiplier of 0.5. Under this calibration, optimal stimulus spending is 7.1 points of GDP, and the unemployment gap is projected to fall by 3.8pp to 2.2pp. Since US GDP in 2009 is $14.7 trillion, optimal stimulus spending is $1.0 trillion. How does this optimal stimulus package compare to the actual stimulus package? The outlays from the American Recovery and Reinvestment Act, enacted into

[2] To obtain public spending as a share of GDP from (18.3), we use the identity $g/y = (g/c)/(1 + g/c)$.

law in February 2009, are estimated to be $663 billion between 2009 and 2019 (CBO 2015, table 1). This is two-thirds of the optimal stimulus spending, but spread over a decade.

18.14. Summary

In this chapter we studied how government spending should be adjusted when unemployment is inefficient. We found that optimal government spending deviates from the Samuelson rule to reduce, but not eliminate, the unemployment gap. The amplitude of the deviation—which corresponds to the amount of stimulus spending—depends on three sufficient statistics: unemployment gap, unemployment multiplier, and elasticity of substitution between public and private goods. Since the unemployment gap is countercyclical, optimal government spending is also countercyclical. That is, the government should spend more in bad times and less in good times.

In the chapter we highlight three main results. The first is that when the unemployment multiplier and unemployment gap are positive, optimal stimulus spending is positive. This is because an increase in public spending reduces unemployment, which is inefficiently high, so it generates a positive effect on welfare that is not accounted for by the Samuelson rule and that warrants raising public spending above the Samuelson rule. The second result is that optimal stimulus spending is zero for a zero multiplier, increasing in the multiplier for small multipliers, maximized for a moderate multiplier, and decreasing in the multiplier for larger multipliers. The monotonicity breaks down because when the multiplier is large, it is optimal to fill the unemployment gap nearly entirely, and then less spending is required to fill the gap when the multiplier is larger. The third result is that optimal stimulus spending is increasing in the elasticity of substitution between public and private consumption. This result is natural: a higher elasticity of substitution means that extra public goods are more valuable, making stimulus spending more desirable.

Given that it is optimal to adjust public spending to the prevailing unemployment gap, it would be beneficial to have an institution that constantly monitors and adjusts public spending, just like there are institutions in place to constantly adjust monetary policy. In the United States, the Federal Reserve constantly monitors the economy so that they can respond quickly to fluctuations in the business cycle. On the other hand, fiscal policy is determined by the Congress and is heavily influenced by political factors. This makes it harder for the legislative branch to act quickly in stabilizing the economy—although fiscal policy, just like monetary policy, should respond to any deviation from full employment. It would be beneficial to develop an agency that adjusts fiscal policy around some average level determined by Congress, so as to stabilize the economy better. The average level of spending ensures that public goods are provided at an appropriate level. The cyclical adjustments allow public spending to pick up the slack in private spending in bad times, and to make room for private spending in good times, which stabilizes the amount of employment and consumption over the business cycle (even though the public/private mix of consumption varies slightly).

In this chapter, we assume that taxation is nondistortionary: it does not affect workers' decisions to participate or not in the labor force. In Michaillat and Saez (2019), we show that the results are unaffected if taxation is distortionary. We introduce an endogenous labor force, modeled as in chapter 8, and we examine how a linear income tax that distorts labor force participation affects optimal government spending. We find that while Samuelson spending is lower with the linear income tax, the correction to the Samuelson rule remains the same. Accordingly, the sufficient-statistic formula developed in this chapter remains valid.

Hence, our theory alleviates a common concern of stimulus skeptics. They worry that output is too low in slumps, and that increasing public expenditure would further reduce output through tax distortions. But our theory shows that if it is only because of tax distortions that public spending reduces output, then stimulus spending should be positive in slumps. Indeed, public spending affects output through two channels: the unemployment channel (public spending reduces unemployment) and the labor-supply channel (more public spending leads to higher taxes, which reduces labor force participation). Then, starting from the Samuelson rule, a small increase in public expenditure reduces unemployment, reduces labor supply, and increases public consumption, which are all good for welfare; but it reduces output and thus private consumption, which is bad for welfare. By construction, at the Samuelson rule, the cost of lower private consumption offsets the benefits of higher public consumption and lower labor supply; the only remaining effect on welfare is the positive effect from lower unemployment. Therefore, increasing public expenditure above the Samuelson rule is desirable. The key is that the Samuelson rule takes into account the negative welfare effect caused by the reduction in output stemming from lower labor supply, but not the positive welfare effect caused by the increase in output stemming from lower unemployment.

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